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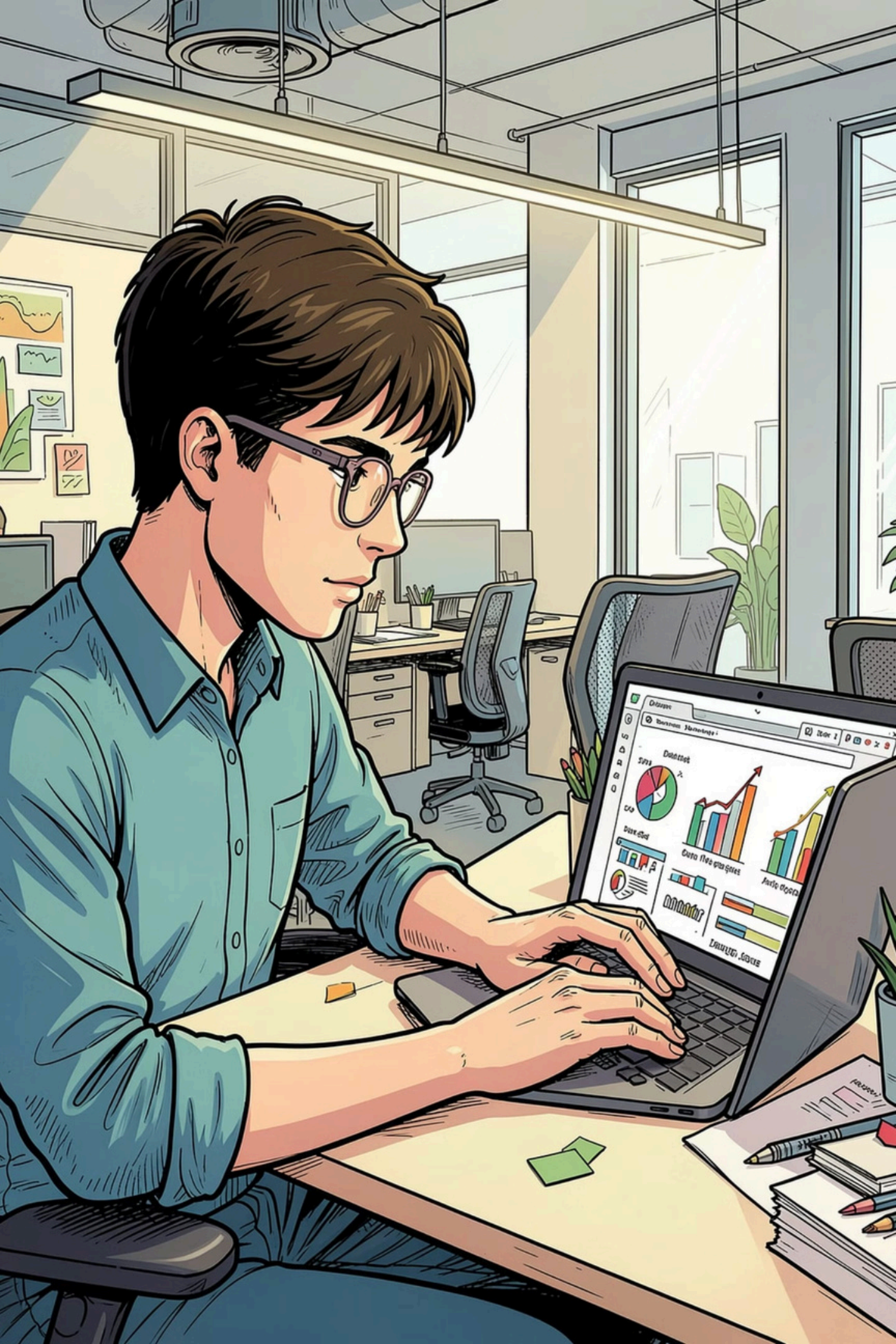
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Sales Performance Analysis Using Python and Power BI

Task 4 - Data Storytelling & Statistical Validation

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Project objective

Analyze an e-commerce sales dataset using Python (data cleaning, EDA, statistical tests) and Power BI (interactive dashboards) to identify customer behavior, sales trends, and business opportunities that can inform operational and marketing decisions.

Business problem

Management needs clear, validated insights to reduce stockouts, increase revenue per customer, and target promotions effectively across cities and age segments.

Dataset overview

Core columns in the dataset:

- Order ID, Order Date, Customer ID, Customer Name
- Age, Gender, City
- Product, Category, Quantity, Unit Price, Total Sales

Data quality notes: Missing values in the **Age** column were filled using the **median**. Missing values in the **City** column were filled using the **mode**.



Exploratory Data Analysis (EDA)



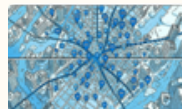
Sales trends

Monthly Sales Trend,
Customer
Demographics, City-
wise Sales, Product
Category Analysis,
Revenue Analysis



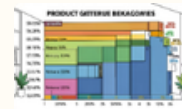
Customer demographics

Age and gender
distribution highlight
primary buyer
segments for targeted
offers.



City-wise sales

Geographic
concentration of
revenue with high-
performing and
underperforming
cities noted.



Product category analysis

Category-level
revenue, margins, and
average order
quantities compared.

Placeholder tiles above for Power BI dashboard screenshots — replace with exported images from the live dashboard.

Key insights (from Power BI)



 **Top Category**
Placeholder — **Electronics**



 **Top City**
Placeholder — **Patna**



 **Top Product**
Placeholder — **Laptop**



 **Highest Sales Month**
Placeholder — **March**



 **Top Age Group**
Placeholder — **46-60**

Statistical validation — Independent Samples T-Test

Purpose: Test whether average order value differs significantly between two independent customer groups (example: Male vs Female).

Hypotheses

- Null hypothesis (H_0): $\mu_1 = \mu_2$ — the group means are equal.
- Alternative hypothesis (H_1): $\mu_1 \neq \mu_2$ — the group means differ.

Test details: Two-sided independent t-test, assumed unequal variances. p-value: 0.49538.

📄 Interpretation: If $p < 0.05 \rightarrow$ reject H_0 (statistically significant difference). If $p \geq 0.05 \rightarrow$ fail to reject H_0 (no significant difference).

Business recommendations



Inventory optimization

Increase inventory for high-selling products from sales trends.
Targeted marketing improve conversion rates and ROI.



Targeted marketing

Run city- and age-segmented campaigns for top products to improve conversion rates and ROI.



Customer loyalty programs

Introduce rewards for repeat buyers and higher lifetime value segments to increase retention.



Seasonal promotions

Plan inventory and discounts around identified high-sales months to maximize revenue capture.



Personalized marketing

Use demographic and purchase-history data for dynamic product recommendations and tailored offers.



SUMMARY RESULTS

- ✓ Checklist completed
- ✓ Checklist, connection
- ✓ Recurrence of data
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- ✓ Checklist click amon

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Conclusion

Project outcomes:

- Comprehensive EDA using Python revealed temporal and geographic sales patterns.
- Power BI dashboards surfaced actionable KPIs: top categories, cities, products, and customer segments.
- Statistical testing (t-test) provided validation for observed differences between customer groups.
- Recommendations focus on inventory, targeting, and promotions to translate insights into revenue.

Value: Empowers ApexPlanet to make data-driven merchandising and marketing decisions that improve sales efficiency and customer lifetime value.

Thank you

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